MAINTENANCE WORK ORDER

Work Order No. : WO-4471
Date Issued : 21 August 2024
Priority : HIGH
Status : COMPLETED
Department : Instrumentation & Electrical
Location : Bay 4, Zone C
Equipment Tag : GB-14 (Draeger Polytron 2 XP Combustible Gas Detector)
Assigned To : Anil Mehta, Senior Technician (I&E)
Supervised By : Rajiv Sharma, Maintenance Supervisor
Approved By : Deepak Singh, Maintenance Engineer

1. WORK DESCRIPTION

Routine inspection and calibration of combustible gas detector GB-14 in Bay 4, Zone C. Work triggered by Shift Handover Log SHL-2024-0819 dated 19 August 2024, which noted GB-14 readings of 18% LEL during shift changeover ? above the 10% LEL notification threshold defined in procedure SHP-04.

A Hot Work Permit (HWP-2024-339) was active in Bay 4 on 19 August 2024 for pipe welding activities at the time of the elevated reading. The welding was suspended at 06:47 IST pending gas reading investigation.

2. WORK PERFORMED

Date Commenced : 21 August 2024, 08:00 IST
Date Completed : 21 August 2024, 11:30 IST

Step 1 ? Bump test performed using 50% LEL methane gas standard.
Response: 47% LEL (within ±5% tolerance). Sensor functional.

Step 2 ? Zero calibration performed. Pre-cal zero reading: +2.1% LEL (drift noted).

Step 3 ? Span calibration performed using 50% LEL certified methane reference gas
(Cert. No. CAL-2024-MET-7723, valid until December 2024).
Post-calibration reading: 50.0% LEL. PASS.

Step 4 ? Zero reading post-calibration: 0.0% LEL. PASS.

Step 5 ? Visual inspection: sensor housing intact. Cable glands secure. No corrosion.
Sensor last replaced: 12 February 2022 (WO-4102). Sensor age: 30 months.
Note: Draeger recommends sensor element replacement every 24 months. OVERDUE.

Step 6 ? Sensor element replacement scheduled under separate work order (WO-4473, pending procurement of Draeger spare ? P/N 4511545).

Step 7 ? Area gas reading post-maintenance: 0.0% LEL. Bay 4 cleared for operations.

3. FINDINGS & OBSERVATIONS

(a) The 18% LEL reading on 19 August 2024 was attributed to a combination of calibration drift (pre-cal zero offset of +2.1% LEL) and actual trace methane present in Bay 4 due to a small flange connection seepage at heat exchanger HX-11 (adjacent to GB-14). HX-11 flange inspection raised separately as WO-4476.

(b) IMPORTANT NOTE: This is the second time in the past 5 years that elevated gas readings on GB-14 have coincided with an active hot work permit in Bay 4. The previous occurrence was documented in Near-Miss Report NMR-2019-047 (March 2019). Recommend this pattern be flagged in the lessons-learned register and that the Hot Work Permit procedure for Bay 4 require mandatory gas reading from GB-14 within 15 minutes (currently 30 minutes per OISD-STD-105).

(c) CA-04 from NMR-2019-047 (digital calibration tracking system) was never implemented. This calibration overdue situation (sensor element 6 months past replacement schedule) would have been prevented by an automated system.